

Digital Transformation of Nepalese Cooperatives Through Aakash DMS

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Abstract

This study examines the digital transformation of Nepalese cooperatives through the implementation of the Aakash Document Management System (Aakash DMS). Nepal's cooperatives have historically faced operational issues, including document mismanagement, slow manual loan processing, compliance gaps, and inefficiencies caused by paper-based workflows. This research explores whether cloud-based automation, digital documentation, and workflow optimization can improve transparency, governance, and operational performance.

The study uses workflow mapping, staff interviews, and performance data from 512 cooperatives across Nepal. The results show major improvements: document retrieval time dropped from hours to seconds, loan processing time was reduced by 50–60%, manual errors decreased by over 80%, and compliance readiness greatly improved. Aakash DMS features include **Digital Know Your Member (KYM)**, automated loan workflows, task management, expiry alerts, and cloud-based archives. These features together cut staff workload by up to 70% and increase accuracy by more than 80%.

The novelty of this study is its sector-level empirical evaluation using real performance data from a diverse network of cooperatives. Results show that Aakash DMS functions as a scalable digital governance model for emerging economies, enhancing efficiency, accountability, and long-term institutional sustainability. Future research should explore artificial intelligence-based risk assessment, optical character recognition for document extraction, and blockchain-based secure archiving.

Keywords: Digital transformation, cooperatives, workflow automation, document management, financial inclusion, Nepal

1. Introduction

Cooperatives in Nepal are among the largest community-based financial networks, providing essential savings, credit, and livelihood support in areas underserved by traditional banks. However, many cooperatives still rely on paper-based systems. Reports from 2023–2025 have highlighted serious governance issues caused by outdated practices, such as missing documents, slow loan processing, high error rates, and difficulties in keeping audit-ready records.

Digital transformation can enhance efficiency, transparency, and sustainability in financial institutions. However, limited large-scale data exists on digital adoption in Nepalese cooperatives. This study evaluates the impact of Aakash DMS—a document management, workflow automation, and compliance tool—by analyzing performance data from over 500 cooperatives.

The study explores the following research questions:

1. How much does digital workflow automation increase operational efficiency in cooperatives?
2. How does centralized digital record-keeping enhance regulatory compliance and audit preparedness?
3. What impact do fewer errors have on operational efficiency, member satisfaction, and institutional trust?

2. Literature Review

International studies indicate that digital governance tools decrease fraud, improve transparency, and increase compliance in financial institutions [1–4]. Cloud-based workflows allow for faster decision-making and keep records accurate and easy to access [5, 6].

However, digital adoption in South Asia faces unique challenges—limited digital literacy, unreliable Internet connectivity, and resistance to technological change [7]. In Nepal, existing studies highlight critical issues such as document mismanagement, inconsistent KYM records, and regulatory non-compliance.

A major gap in the literature is the absence of large-scale, data-driven research on digital transformation in Nepalese cooperatives. Most current studies are either qualitative or small-scale. This research addresses that gap by examining long-term logs from various cooperatives using a standardized digital management system.

3. Materials and Methods

3.1 Study Design

This study employed a mixed-methods **quasi-experimental design**. Quantitative system log data were analyzed through a **pre–post comparison**, examining key performance indicators (KPIs) six months before and after the implementation of Aakash DMS. Qualitative interviews were conducted to interpret the quantitative findings, identify challenges in adoption, and collect user experiences.

3.2 Ethical Statement

Participating cooperatives signed official agreements granting access to anonymized system data. All interviews were conducted with informed consent, and no personally identifiable information was collected.

3.3 Sampling Criteria

A total of 512 cooperatives were selected using a purposive sampling method to ensure data reliability, diversity, and representativeness. The selection was based on the following criteria:

- 1. Minimum Digital Adoption Period:**

Each cooperative must have actively used Aakash DMS for at least six months to ensure reliable post-implementation performance data.

- 2. Operational Scale Requirements:**

Cooperatives had to meet a minimum operational threshold of either:

- $\geq 2,000$ active members, or
- ≥ 500 monthly financial transactions.

This provided enough data for meaningful analysis.

- 3. Data Access and Participation Commitment:**

Only cooperatives willing to provide anonymized system logs and participate in structured interviews were included in the study.

4. Geographical Representation:

To reflect differences in infrastructure, digital literacy, and operating environments, samples were gathered from both the Terai and Hill regions of Nepal.

5. Institutional Diversity:

The sample included different types of cooperatives, such as Savings and Credit Cooperatives (SACCOS) and Multipurpose Cooperatives, to ensure comparability across sectors.

3.4 Data Validation and Analysis Protocol

- **Document Retrieval Time:** the period from staff query to file opening, as logged in the system.
- **Loan processing time:** from application submission to final disbursement.
- **Error Rate:** determined by comparing 50 pre-DMS physical loan files and 50 post-DMS digital entries to official member records.
- **Operational Workload:** evaluated through time-motion studies and staff interviews, quantifying hours spent on administrative tasks before and after DMS implementation.

4. Results

Data collected from 512 cooperatives substantially reveal significant operational gains following the implementation of Aakash DMS.

4.1 Operational Efficiency

Document retrieval time decreased by over 99%, dropping from 2–5 hours to just 5–20 seconds. Similarly, total loan processing time was reduced by about 50–60%, going from 3–7 days to 1–2 days.

4.2 Error Reduction

Manual error rates in key processes dropped from 40–60% to 5–10%, marking an improvement of nearly 80–85%. This reduction has resulted in increased data accuracy and more reliable record-keeping.

4.3 Compliance and Workload Optimization

Compliance readiness increased significantly due to automated documentation and system-generated reminders. Staff administrative workload decreased by nearly 70%, allowing personnel to dedicate more time to member services, field visits, and strategic planning.

Table 1: Operational Improvements After Aakash DMS Implementation (N=512)

Metric	Pre-DMS Baseline	Post-DMS Measured	Improvement / Change
Document Retrieval Time	2–5 hours	5–20 seconds	>99% reduction
Loan Processing Duration	3–7 days	1–2 days	50–60% reduction
Manual Error Rate (Est.)	40–60%	5–10%	80–85% reduction
Compliance Readiness	Low	High	Substantial increase
Operational Workload	100%	~30%	~70% reduction

5. Discussion

Digital workflows directly address key operational challenges faced by Nepalese cooperatives. They decrease retrieval times and errors, enhancing transparency and trust within the organization. A 70% reduction in administrative tasks enables staff to focus more on member support and strategic planning.

5.1 Novelty and Contribution

This study is distinctive because it evaluates **sector-wide digital transformation** using real, large-scale operational data. Prior research in South Asia mainly comprises small case studies or literature reviews. The findings demonstrate that Aakash DMS provides a scalable governance model suitable for environments with unreliable connectivity.

5.2 Challenges and Limitations

Challenges included limited digital literacy among older staff and inconsistent Internet connectivity in remote areas. Training and system design helped address these issues. Although this study offers strong pre–post evidence, future research should explore long-term effects on member trust, retention, and financial performance.

6. Conclusion

Aakash DMS effectively addresses long-standing governance and operational challenges in Nepalese cooperatives. Transitioning from paper-based methods to automated cloud workflows significantly boosts efficiency by reducing processing times, minimizing manual errors, and enhancing regulatory compliance. These improvements demonstrate that digital transformation can play a crucial role in strengthening accountability and service delivery within the cooperative sector.

The findings further develop indicate that the Aakash DMS model is scalable and well-suited for cooperatives operating in similar socio-economic and infrastructural contexts. Future advancements could include artificial intelligence for credit risk assessment, optical character recognition (OCR) to digitize historical documents, and blockchain technology for secure and permanent record-keeping, supporting long-term digital governance and institutional sustainability.

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8. References

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